

DEPRESSION AND LONELINESS IN OVERSEAS STUDENTS

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SUMMARY

This paper compared the effect of life change in adjustment of overseas students with those of Australian students. 342 students were divided into 4 experimental groups. These were: Overseas students with residence of less than one year (OV1 group, N = 44), overseas students with residence of more than one year (OV2 group, N = 81), Australian students raised in Brisbane (A3, N = 105) and Australian students raised outside Brisbane (A4, N = 112). The result of discriminate analysis showed that overseas students (OV2) were significantly more likely to experience moderate to severe clinical depression and loneliness than the Australian students. Age and expectation of difficulty in study were the best predictors for depression in overseas students.

INTRODUCTION

Culture shock is a term used to describe the anxiety that results from the loss of one's sense of wellbeing as a result of the removal or distortion of many of the family cues in the home country (Locke & Feinsod, 1982). Given these cultural differences, and misunderstandings of the new and diverse experiences, feelings of alienation, anxiety and depression are likely to be experienced by people who travel or migrate to foreign countries, whether permanently or temporarily (such as overseas students) as part of the 'transitional experience' or adjustment process (Alder, 1975; Masuda, Lin & Tazuma, 1982; Karamanos, 1982; Nguyen, 1982; Bradley & Bradley, 1985; Syed, 1983).

The course of adjustment of sojourners over time has frequently been depicted by a U-curve pattern (Church, 1982). That is an early period of curiosity, expectation and enjoyment, followed by a period of disappointment and disillusionment, as the sojourners become aware of the demands of the new culture. Gradually the sojourners learn to cope with the demands, and reach a 'coming to terms' phase when they start establishing satisfying personal relationships and build up their levels of self-esteem. The U-curve hypothesis received inconsistent support (see Church, 1982). Becker (1968), for example, reported that the U-curve operated in reverse pattern for the students who came from semi- or under-developed countries. Church (1982) claimed that the U-curve pattern occurs only in a small minority of cases.

In recent years there has been a growing recognition of the differences in patterns of adjustment according to individual differences, conditions in the host country, prejudice towards immigrants, the characteristics of the immigrants themselves and their pre-existing vulnerability

(Verdonk, 1979; Stening, 1979). The greater the differences between the background culture and the host culture, the more stressful the sojourners' stay (Graham, 1983; Stening, 1979).

In terms of individual differences, certain predispositions seem to make a person more vulnerable to stress and anxiety. For example, Ellis (1962), strongly argued that irrational beliefs caused dysfunctional emotion. Spielberger (1966) reported that anxiety correlated highly with the level of stress individuals experienced. Among these traits that predispose travellers to debilitating culture shock are depression, anxiety, obsessive-compulsive traits, psychosomatic and hypochondriacal predispositions and authoritarian character (Locke & Feinsod, 1982). Individuals' cognitive styles are other determinants of the level of culture shock experienced. Detweiler (1980) reported that people who have broad category width in their cognitive structure are more tolerant to cultural differences compared to those with narrow category width.

Besides cultural and cognitive differences, demographic variables such as age, marital status and educational level are also reported as important factors in determining the quality of intercultural adjustment and the presence of psychopathology (Okamura & Collier, 1978; Pruitt, 1978; Jammaz, 1972; Levitt, Lubin & Brooks, 1983). Age on arrival in the host country generally correlates negatively with overall assimilation. Younger students seem to adjust more easily than the older ones, female students are less adjusted than males, and married students (without their spouses) are less adjusted than single ones (Pruitt, 1978).

In summary, previous studies have pointed out that studying in a foreign country often brings about a considerable level of stress, anxiety and depression. The level of the overseas students' emotional problems seems to vary according to their length of stay, their personality traits and their demographic status (e.g., age and sex). However, there is also contradictory evidence against the particularity of the overseas students' adjustment problems; for example, it has been reported that between 10% to 20% of freshmen complained regularly of emotional symptoms (Bradley *et al.* 1985). This study intended to investigate the psychological problems of first year overseas students as compared to the indigenous first year Australian students. This comparison would allow us to evaluate whether the overseas students' adjustment problems were particularly due to new academic environment (as experienced by the Australians raised in Brisbane), or to new academic environment and new habitat (as experienced by Australians raised outside Brisbane), or to the overall new culture. Based on personality and demographic variables, the predictability of depression was also investigated.

METHOD

Subjects

342 subjects (125 overseas and 217 Australian students) participated in this study. Most of the overseas students (84%) came from Asian countries, such as Malaysia and Singapore. The Australian students were recruited through the first year psychology student pool. They participated in the study to gain some experimental hours. Length of stay in the area (Australia) was considered to be an important predictor in the adjustment level, and thus the overseas students were differentiated into two groups: those who had been in Australia for less than one year (OV1), and those who had been here for more than one year (OV2); and the Australians were differentiated to those raised in Brisbane (A3), and those raised outside of Brisbane (A4). There were 44 overseas students who had been in Australia for less than one year, 81 overseas students

who had been here for more than one year, 105 Australian students who had been raised in Brisbane, and 112 Australian students who had been raised outside of Brisbane.

Apparatus

The psychological scales in this study were:

Beck Depression Inventory (BDI). The BDI is a 21 item self-report inventory designed to assess clinical depression (Beck *et al.* 1961).

Loneliness Scale (LS). This is a 20-item self-report rating scale measuring feelings of loneliness (Russell *et al.* 1972).

Irrational Beliefs Tests (IBT). This test is a 100 item self-report questionnaire. It was constructed to measure the 10 irrational beliefs (Ellis, 1962).

Trait Anxiety Inventory (STAI). The inventory was part of the STAI. It consisted of 20 statements that ask respondents to describe how they generally feel on a 4 point likert scale. For each individual self-report questionnaire, a total score can be obtained by summing across all items. The greater the score, the greater the severity of the variable. A total score ranging from 0 to 63 is obtained. The greater the score, the greater the severity of depression. In addition to these scales, the students were asked to rate their overall adjustment level and activities. The overseas students were also requested to rate their academic achievement in their home countries, their difficulties with study before coming to Australia, their expectations of difficulties here, and their difficulties in social relationships and English proficiency using visual analog scale.

PROCEDURE

Overseas students

The survey of overseas students was conducted either directly or indirectly by one of the researchers. In direct contact, the researcher herself distributed and collected questionnaires individually. In indirect contact, an intermediary carried out the distribution and collection of the questionnaires. The participants, all volunteers, were encouraged to read through the questionnaire in the presence of the researcher or the intermediary. Any questions that were ambiguous to the subjects were explained. The questionnaires were then left with the participants for completion and on average, were collected one week later.

Australian students

The Australian students completed the questionnaires collectively at appointed times in a classroom in the Psychology Department. After a brief introduction and explanation of the study, the questionnaires were distributed for completion.

In total 403 questionnaires were distributed, 370 (92%) of them were returned. However, only 342 (85%) were analysable.

RESULTS

The data analysis for this paper was based on information from 342 subjects. Of the 370 returned questionnaires, 28 subjects were rejected because the questions were not completed properly.

Prior to the analysis of the data, examinations were made for relevant multivariate statistical assumptions concerning normality and multicollinearity of the rating scales. The results showed that no violation of assumptions was detected. Discriminate analysis was therefore used to examine the differences between the four groups of students, and multiple regression was used to examine the power of the demographic and personality characteristics in predicting depression.

Table 1 shows the personality variables used in the discriminant analyses to investigate whether any of the 14 variables could be used to discriminate between the 4 sample groups. The results of the discriminant analysis showed both function 1 ($\lambda = 83.68$, $df 16$, $p < 0.0001$) and function 2 ($\lambda = 22.06$, $df 144$, $p < 0.0107$) were significant and accounted for 68.8% and 22.2% of the variance respectively. The loadings of individual variables with 0.4 or greater for functions 1 and 2 are shown in Table 1.

TABLE 1
Standardized Canonical Discriminant Functions Coefficients
for the Overseas and Australian Students

	Function 1	Function 2
1. Loneliness	—	.4
2. Anxiety Trait	—	—
3. Depression	—	.4174
4. Demand for Approval	—	—
5. High Self-Expectation	—	—
6. Blame Proneness	—	—
7. Frustration Reactivity	—	—
8. Emotional Irresponsibility	—	—
9. Anxious Overconcern	—	—
10. Problem Avoidance	-.4491	—
11. Dependency	-.4955	—
12. Helplessness	—	—
13. Perfectionism	.5947	—
14. Overall Adjustment	—	—
Percent variance	68.8	22

The results show that problem avoidance, dependency and perfectionism scored highly on function 1, and that loneliness and depression loaded significantly on function 2. Figure 1 shows that the Australian students (groups A3 and A4) cluster together, and can be distinctly separated from the overseas groups (groups OV1 and OV2) on function (i.e. irrational belief factors). Similarly on function 2, two clusters were formed: overseas students who had stayed in Australia up to one year (OV1) and the Brisbane students (A3) formed one cluster, and overseas students who had been in Australia for more than one year (OV2) and the rural Australians (A4) formed the other cluster. Univariate F-tests for all variables are also presented in Table 2. Examination

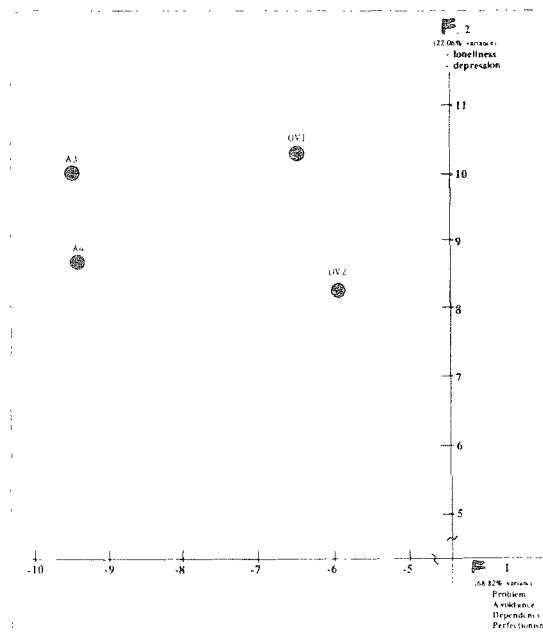


Figure 1.

Groups centroid for the four experimental groups. OV1 = Overseas Students with less than one year of residency in Australia, OV2 = Overseas students with more than one year of residency in Australia, A3 = Australian students raised in Brisbane, and A4 = Australian Students raised outside Brisbane.

of the results reveals that the variables which contributed significantly were depression ($p < 0.01$), frustration reactivity ($p < 0.05$), problem avoidance ($p < 0.01$), dependency ($p < 0.01$), helplessness ($p < 0.05$), and perfectionism ($p < 0.0001$). Analysis with Newman-Keuls reveals that only the frustration reactivity variable can discriminate between the two overseas student groups (OV1 and OV2). The tendency to catastrophize adversities (frustration reactivity), as well as the tendency to dwell on fearsome matters (anxious overconcern) of the overseas students who had been here for less than one year, were also significantly less than those of the Australians (A3 and A4 groups). Dependency and need to seek perfect solutions (perfectionism) were the variables that significantly discriminated both overseas student groups and the Australian groups. The Australians also had a significantly higher tendency to avoid problems compared to overseas students who had been here for less than one year (OV1), but compared to overseas students who had been in Australia for more than one year (OV2) the difference was not significant. On the other hand, overseas students who had been here for more than one year (OV2) had significantly higher feeling(s) of helplessness (i.e. one is a helpless victim of the past) compared to the Australian groups. Overseas students who had been here for more than one year were the most depressed, when compared with the Australian groups.

Further analysis of the distribution (see Table 3) of the depression scores between the Australian and overseas groups revealed that 53.6% of overseas students experienced moderate to severe levels of depression as compared to the 22.3% of rural Australians (A4) and 16.2% of the Brisbane-Australians (A3). Furthermore, when clinical depression criteria was used (i.e. BDI score < 22), it was found that 10.4% of overseas students were more likely to be clinically depressed as compared with 4.46% rural-Australians and 1.9% Brisbane-Australians. Comparisons between pairs of groups using chi-square statistics revealed that the percentage of overseas students' depression scores was significantly different from the Brisbaners' (χ^2 1,230 = 44.14, $p < 0.001$) and the rural-Australians (χ^2 , $df = 1 = 16.5232$, $p < 0.01$). The rural-Australians' depression scores were not significantly different from the Brisbaners'.

Multiple regression of overseas students

Since overseas students were more likely to experience depression, it was decided to investigate the predictability of overseas students' depression from their demographic variables (see Table 4). The results indicate that depression scores can be predicted significantly by the background and demographic variables (see Table 4) (R square = 0.2927, F -(18.106) = 2.437, $p < 0.01$). Table 4 presents the regression coefficients of the variables, the standardised coefficients (beta) and the unstandardised ones (b). It was found that age and expectation of study difficulties were the only significant predictor variables.

TABLE 2
The means and univariate f-test for the four groups

	F-ratio	p	Group means			
			OV2	OV1	A3	A4
1. Loneliness	1.9399	.1214	37.7045	39.6420	39.4286	36.1810
2. Anxiety Trait	.0280	.9932	43.4091	43.0741	43.4643	43.3428
3. Depression	4.1560	.0068	7.6136	9.7778	7.1786	6.1238
4. Demand for Approval	1.5929	.1893	15.3182	15.2469	16.2411	15.6762
5. High Self-Expectation	.6863	.5645	16.4773	16.4321	15.9375	16.4857
6. Blame Proneness	.6586	.5818	15.9318	16.2099	15.8393	15.4672
7. Frustration Reactivity	2.8778	.0353	15.7045	16.8889	16.8750	16.9524
8. Emotional Irresponsibility	.5790	.6334	13.4091	13.0864	13.4821	13.0762
9. Anxious Overconcern	3.8398	.0101	14.4318	15.3827	16.3125	15.7048
10. Problem Avoidance	5.8198	.0010	12.8409	13.4568	15.0268	14.4000
11. Dependency	6.6949	.0004	15.8864	16.3704	17.6607	17.8190
12. Helplessness	3.4848	.0160	14.1818	15.0000	13.8036	13.6571
13. Perfectionism	9.6973	.0000	14.7500	14.1605	12.4107	12.4381
14. Overall Adjustment	1.3251	.2650	5.7500	6.1605	6.3393	6.2476

degrees of freedom: dfb: 3 dfw:338

TABLE 3
Depression scores categories among the groups

	Nondepressed $x < 11$ N (%)	Moderately Depressed $12 < x < 21$ N (%)	Severely Depressed $x < 22$ N (%)
Overseas	58 (46.4)	54 (43.2)	13 (10.4)
Rurals	87 (77.68)	20 (17.8)	5 (4.46)
Brisbaners	88 (83.81)	15 (14.2)	2 (1.9)

TABLE 4
**Regression coefficients of overseas student demographic variables
in predicting depression**

	beta	b
1. Study difficulties expectation (Australia)	.2431	2.0572*
2. Age	-.1885	-.3078*
3. Academic expectation	-.1431	-1.5015
4. Academic achievement (home)	-.1313	-.8847
5. Length of stay in Australia	.1209	.6760
6. Difficulties with study (home)	.1033	.9631
7. Activities membership	.0974	1.4678
8. Brothers/sisters in Brisbane	.0871	1.8144
9. Marital status	-.0744	-1.7819
10. Language proficiency (self)	-.0622	-.4813
11. Social difficulties (Australia)	.0620	.5574
12. Other relatives in Brisbane	.0490	1.4198
13. Level of study	-.0355	-.1698
14. Church attendance	.0332	.1511
15. Sex	-.0310	-.5135
16. Language proficiency (judged by others)	-.0240	15.0339
17. Length of stay in Brisbane	-.0095	-.0553
18. Parents in Brisbane	.0000	.0000

* $p < .05$

DISCUSSION

The major findings indicate that overseas students who had been here longer than one year were more likely to experience depression than Australian-Brisbaners. While loneliness and overall adjustment ratings do not lend support to the hypothesis that the greater the life change the more

adjustment problems a person has, the order of depression levels of the four groups was consistent with the prediction. It seems that the greater the environmental changes (Brisbaners experienced the least and overseas students experienced the most) the greater the level of depression. The different courses of loneliness and depression that were experienced over time by the overseas students suggests that the overseas students' adjustment process involves multidimensional problems.

One possible explanation of the presence of a significant difference in depression but an absence in loneliness, is the nature of friendship and the level of coping skills developed by the overseas students. It is often suggested that foreign students in their study reported no close host-national friends at all (Furnham & Alibhair, 1985). Most of them reported having a co-national best friend. Church (1982) pointed out that while these co-national social enclaves restricted their interaction with the host nationals, it allowed the overseas student to establish intimate relationships and to restore a sense of belonging and self-esteem. In other words, the enclaves could reduce the feelings of loneliness, but not necessarily the feelings of depression. However, as most of the overseas students' friends are likely to be other overseas students, their direct contacts are likely to be limited to the length of their stay in the host country. While they study and share their experiences with each other, the friendship provides mutual support. However, once they finish their studies and go home, those left behind would naturally lose their direct emotional support and would feel lonely. Over time the amount of friends could be predicted as decreasing and the level of loneliness as increasing. On the other hand, as their coping skills become better, the overseas students seem to be less depressed.

The depression pattern adds to the U-curve hypothesis of the overseas students' adjustment, but the loneliness pattern does not. A check on the self-rating overall adjustment pattern over period of stay does not confirm the U-curve pattern either. It implies that the course of overseas students' adjustment is not unidimensional.

Fifty-four percent of overseas students experienced moderate to severe depression (Table 3). This is definitely a concern. It is possible that this factor may hinder the process of adjustment and progress of academic studies for these students. A report from the University of Queensland counselling service confirmed the seriousness of the problem. In 1982, 8.9% of the individual counselling interviews were conducted with the overseas students, who represented about 3% of the student population (Student Services, 1982, Committee of Review, 1983).

Predicting the overseas students' depression

The results suggest that expectation of difficulty in study and the students' age were the most powerful predictors of overseas students' depression scores. A positive regression coefficient and high positive bivariate correlation between the expectation of difficulty concurs with high level of depression. This raises the question of whether it is the expectation of academic problems which precipitated the depression. Rao's (1979) survey revealed that the overseas students tended to perceive more problems than they actually experienced. Research so far shows that cognitive processes have a great impact on an individual's emotion (Ellis, 1962). Expectation of academic problems reported correlated significantly with self-rated language proficiency, self-expectation of academic achievement, activities participation, previous academic difficulties and previous academic achievement. Those who had difficulties in study before and those who had poor language proficiency seemed to expect more academic problems, and reported greater

depression. Considering the nature of the academic tasks, high correlations of academic problems with these variables is understandable. Since studying is the main purpose of their coming to the host country, it is also understandable that their self-prediction of academic problems or failure would bring about their depression.

Age was another significant predictor to depression scores. The regressive coefficient between age and depression scores were negative, indicating that the older the overseas students, the less depressed they were. This result disagrees with the results of Levitt *et al.* (1983), which showed that age correlated positively with depression. There is no apparent reason for this disagreement. A possible explanation is that the differences could be due to differences in the samples and the countries with which these studies were carried out.

The present findings show that the Australian and overseas students could not be differentiated on anxiety traits. However, there were significant differences between them in some irrational belief variables. A comparison between the overseas and the Australian students showed that the Australians had a significantly higher tendency to rely on others (dependency), but a significantly lower tendency to seek a perfect solution (perfectionism). Furthermore, the overseas students who had been here less than one year were also less easily frustrated (less frustration reactivity), less worried about future misfortunes (less anxious overconcern), and less prone to avoid problems (less problem avoidance) as compared to the Australian students. These results therefore confirm the earlier suggestion that the more recent visitors were more enthusiastic and more relaxed. This confirmed the hypothesis that at their initial period of stay, overseas students were excited, full of curiosity, expectation and enjoyment of the novel (Livingstone, 1960). These attitudes were less prominent in the overseas students who had been here for more than one year and this may also account for some of the depression experienced by them.

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